ALAO JEREMIAH

Email: timix295@gmail.com | Phone: +2349050772646 | Location: Ikorodu, Lagos, Nigeria |
LinkedIn: linkedin.com/in/jeremiah-alao | GitHub: github.com/youniquestar |
Portfolio: youniquestar.github.io

RESEARCH INTERESTS

First-class Mechatronics Engineering student specializing in vision-assisted robotic manipulators, control theory, and haptic feedback. Seeking to pursue advanced graduate research in autonomous physical manipulation and robotics control systems.

EDUCATION

Bachelor of Engineering (B.Eng.) in Mechatronics Engineering
Lagos State University of Science and Technology, Nigeria

- **Expected Graduation:** July 2027
- **Class Rank:** First Class Honors (Top 10% in Department)
- **Key Coursework:** Robotic Kinematics, Modern Control Systems, Embedded Microcontrollers, Computer Vision, Linear Algebra, Dynamic System Modeling.

RESEARCH EXPERIENCE & ACADEMIC PROJECTS

Design of a 5-DOF Vision-Assisted Robotic Arm with Haptic Feedback
Lead Undergraduate Researcher

- Engineered and fabricated a 5-DOF robotic arm utilizing Autodesk Fusion 360, integrating dynamic torque modeling and physical kinematic constraints.
- Implemented closed-loop motor control and trajectory planning algorithms in C++ and ROS/ROS2, bridging CAD physical modeling with mathematical constraints.
- Developed a visual perception pipeline using OpenCV to translate visual inputs into physical grasp execution.
- Integrated haptic feedback and programmed embedded microcontrollers (STM32/ESP32) to facilitate tactile sensing and real-time system debugging.

Autonomous Obstacle-Avoidance Mobile Robot
Lead Systems Integrator & Developer

- Designed and built an autonomous mobile robot platform utilizing Arduino, L298N H-bridge motor drivers, and high-discharge Li-Ion power supply rails.
- Engineered a reactive navigation state-machine algorithm in C++ incorporating multi-angle ultrasonic scanning, median filtering, and differential-drive maneuvers to navigate dynamic environments.
- Fabricated custom mechanical brackets and panning sensor turrets in Autodesk Fusion 360 and 3D printed them using FDM technology (Elegoo Neptune 3 Pro).

Human Following Autonomous Robotic Carriage
Hardware & Controls Developer

- Developed an autonomous target-tracking robotic carriage employing an array of ultrasonic and infrared sensors for target tracking and obstacle avoidance.
- Programmed real-time control algorithms in C++ to process sensor telemetry, enabling smooth target tracking and responsive collision avoidance.

TECHNICAL & LABORATORY SKILLS

- **CAD & Simulation:** Autodesk Fusion 360, MATLAB/Simulink, Gazebo
- **Programming & Frameworks:** C++, Python, ROS / ROS2, OpenCV, Linux
- **Hardware & Microcontrollers:** STM32, Arduino, ESP32, Embedded C
- **Prototyping & Equipment:** 3D Printing, Closed-loop Actuators, Sensor Integration, Soldering

HONORS & AWARDS

First Class Honours Standing – Ranked in the Top 10% of the Mechatronics Engineering graduating cohort.